RESEARCH SYNOPSIS

PART A: PERSONAL DETAILS

1. **Name of the Principal Investigator:** Mr. Joju
2. **PI's Designation and Name of the Department and College/Centre:**
    - **Designation:** 1st Year Postgraduate Student
    - **Department:** Speech-Language Pathology
    - **College/Centre:** Bangalore University, Bengaluru, Karnataka
3. **Name of the Course:** MSc Speech-Language Pathology
4. **Date of admission to course:** [To be provided by the candidate]
5. **Contact details of the PI:**
    - **Valid mobile number:** [To be provided by the candidate]
    - **Active email id:** [To be provided by the candidate]
6. **Name(s) of the PG Guide/Co-guides/Co-investigators/Research team members:**
    - **Guide:** Mr. Hemaraja Nayaka.S, Associate Professor, Department of Speech-Language Pathology, Yenepoya Medical College Hospital, Mangalore
    - **Co-Guide:** Dr. Anne Varghese, Professor and Principal, NAISH, Bangalore University

PART B: STUDY DETAILS

1. TITLE OF THE RESEARCH TOPIC:

Effectiveness of AI-Developed Gamified Virtual Reality Simulations Using Surface Electromyography (sEMG) Biofeedback Compared to Conventional Therapy on Swallowing Function in Adults with Post-Stroke Neurogenic Dysphagia: A Prospective Randomized Controlled Trial.

2. EXECUTIVE SUMMARY:

Post-stroke dysphagia (PSD) is a highly prevalent complication that significantly impairs patients' quality of life and leads to severe consequences such as aspiration pneumonia and malnutrition. While surface electromyographic biofeedback (sEMG-BF) and virtual reality (VR) have independently shown promise in rehabilitation, custom clinical software is often cost-prohibitive. This study pioneers the rapid prototyping of a targeted "anti-gravity" digital simulation using modern AI-assisted development frameworks specifically, the Cursor IDE and the Antigravity agentic platform. The simulation uses sEMG input to control a gamified cursor, providing real-time visual biofeedback. A prospective randomized controlled trial will be conducted with 30 post-stroke patients (15 in the experimental group, 15 in the control group) to determine whether this customized, gamified VR intervention yields statistically significant improvements in the Functional Oral Intake Scale (FOIS) and patient adherence when compared to standard behavioral swallowing therapy over a 6-week period.

3. INTRODUCTION & BACKGROUND:

Stroke is a leading cause of oropharyngeal dysphagia, affecting nearly 50% of stroke survivors. Standard dysphagia rehabilitation heavily relies on behavioral maneuvers, such as the effortful swallow and Mendelsohn maneuver. However, these conventional exercises are fundamentally abstract, often leading to poor patient adherence due to the lack of real-time sensory feedback. Swallowing function recovery depends heavily on the neuroplastic development of the motor cortex, a process that can be accelerated through auditory and visual stimulations. Surface electromyography (sEMG) captures the electrical activity of swallowing muscles, allowing for real-time visual feedback that augments a patient’s self-awareness and control over swallowing movements. Recently, immersive virtual reality (VR) and gamification have been integrated into stroke rehabilitation to provide interactive, feedback-rich environments that promote active participation and sustained motivation.

4. NEED FOR THE STUDY:

Despite the known benefits of VR and sEMG biofeedback, the development of customized digital environments for dysphagia therapy has traditionally required extensive software engineering resources, making it unfeasible for most speech-language pathology departments. There is an urgent need to democratize the creation of personalized clinical rehabilitation tools. The advent of AI-driven integrated development environments (IDEs) like Cursor and agent-first development platforms like Google's Antigravity allows clinicians to rapidly prototype such software. This study addresses a dual gap in the literature: it tests a novel methodological

approach to clinician-led software development, and it assesses the clinical efficacy of the resulting custom anti-gravity sEMG simulation in improving functional swallowing outcomes.

5. RESEARCH QUESTION:

a. RESEARCH QUESTION OR HYPOTHESIS

- **Research Question:** Does the adjunctive use of a custom-developed, gamified VR simulation (built using Cursor and Antigravity) utilizing sEMG biofeedback improve functional oral intake and swallowing-related quality of life more effectively than conventional swallowing therapy alone in patients with post-stroke dysphagia?
- **Null Hypothesis (H0):** There is no significant difference in the Functional Oral Intake Scale (FOIS) scores between patients receiving custom VR sEMG-biofeedback therapy and those receiving standard conventional therapy.
- **Alternate Hypothesis (H1):** Patients receiving custom VR sEMG-biofeedback therapy will demonstrate significantly higher FOIS scores compared to those receiving standard conventional therapy.

b. AIM To develop a custom VR dysphagia simulation using AI-assisted platforms and to evaluate its clinical effectiveness on swallowing function and patient engagement in adults with post-stroke dysphagia.

c. OBJECTIVES

1. To engineer and deploy a functional gamified simulation using Cursor and Antigravity where submental sEMG peak amplitude directly controls a visual anti-gravity cursor.
2. To compare the pre-intervention and post-intervention swallowing function between the experimental and control groups using the Functional Oral Intake Scale (FOIS).
3. To evaluate the impact of gamified VR therapy on patient adherence and swallowing-related quality of life using the Swallowing Quality of Life Questionnaire (SWAL-QOL).

6. REVIEW OF LITERATURE:

Nordio et al. (2022) conducted a pilot-randomized controlled trial on 17 patients to analyze the effectiveness of sEMG biofeedback as an adjunctive treatment for post-stroke dysphagia. While the primary outcome, the Functional Oral Intake Scale (FOIS), improved across all patients, the sEMG-biofeedback group demonstrated significant specific improvements in pharyngeal clearance and swallowing safety, effectively reducing the risk of aspiration.

A recent systematic review and network meta-analysis by Liu et al. (2025) evaluated the efficacy of sEMG biofeedback compared to conventional therapy. Analyzing data from six randomized controlled trials, they found that sEMG-biofeedback significantly increased mean swallowing amplitude and improved Standardized Swallowing Assessment (SSA) scores. The meta-analysis concluded that sEMG-biofeedback outperforms conventional therapy by enhancing patients' self-awareness and control over swallowing movements, although the overall impact on SWAL-QOL did not reach statistical significance, indicating a need for more engaging modalities.

To target engagement directly, a meta-analysis of randomized controlled trials (2026) published in *JMIR Serious Games* investigated the effectiveness of gamified swallowing exercises. The review synthesized data from 330 post-stroke patients and demonstrated that incorporating game-like elements (e.g., real-time visual rewards) alongside sEMG-biofeedback significantly improved overall swallowing performance (Mean Difference = 1.1; 95% CI 0.9 to 1.3). Gamification directly augmented patient motivation and adherence, factors historically lacking in conventional behavioral therapy.

Furthermore, Wu et al. (2024) explored the impact of immersive VR technology on stroke patients with swallowing disorders. In a study of 100 patients, the experimental group receiving VR-based visual and auditory stimulation showed a functional improvement rate of 94.23% and a patient satisfaction rate of 88.46%, significantly higher than the conventional therapy group.

Collectively, these studies affirm the clinical value of sEMG biofeedback and VR in dysphagia rehabilitation. However, there is no existing literature detailing the autonomous development of these tailored interventions by allied health professionals using generative AI coding platforms like Cursor and Antigravity.

7. METHODOLOGY:

a. STUDY DETAILS:

i. STUDY DESIGN: Prospective, interventional, parallel-group, randomized controlled trial.

ii. STUDY SITE: Outpatient rehabilitation clinics and the Department of Speech-Language Pathology at Bangalore University, Bengaluru.

iii. FUNDING DETAILS: Self-funded by the principal investigator.

iv. STUDY DURATION: 8 months.

b. PARTICIPANT DETAILS:

i. SOURCE OF DATA: Post-stroke patients attending the neurology and rehabilitation departments affiliated with Bangalore University.

ii. SAMPLE SIZE: A sample size of 30 participants (15 per group) is required. This is calculated based on the *JMIR Serious Games* (2026) meta-analysis, which reported an expected mean difference of 1.1 in functional swallowing scores with a standard deviation of 1.2. With a 95% confidence interval (alpha = 0.05) and 80% power, the calculated sample size is 20. To account for a 20-30% attrition rate typically seen in post-stroke populations, the total sample size was padded to 30.

iii. SAMPLING METHOD: Consecutive purposive sampling followed by computer-generated simple randomization.

iv. INCLUSION CRITERIA: 1. Adults aged 45 to 75 years. 2. Confirmed diagnosis of neurogenic oropharyngeal dysphagia secondary to a stroke. 3. Adequate cognitive functioning to follow 2-step verbal instructions and interact with a screen.

v. EXCLUSION CRITERIA:

Structural or mechanical dysphagia (e.g., head and neck cancer history).

Severe visual/auditory impairments preventing the use of screens.

c. INTERVENTION:

Phase 1 (Software Development): The investigator will utilize Cursor (AI IDE) and Antigravity to code an "anti-gravity" visual simulation. The software will interface with a clinical sEMG unit. When the patient executes an effortful

swallow, the submental sEMG sensors capture the electrical amplitude. The software interprets this to lift a virtual object against simulated gravity.

Phase 2 (Clinical Trial): * *Control Group (n=15):* 45 minutes of conventional behavioral swallowing therapy (e.g., effortful swallow without biofeedback) conducted by an SLP, 3 times a week for 6 weeks.

- *Experimental Group (n=15):* 45 minutes of the VR sEMG-biofeedback therapy, 3 times a week for 6 weeks, controlling the custom anti-gravity simulation using swallowing efforts.

d. DATA ANALYSIS: Data will be analyzed using SPSS Version 28.0. An Independent Samples t-test will compare the mean differences in FOIS and SWAL-QOL scores between the control and experimental groups post-intervention. A Paired Samples t-test will assess within-group changes. A p-value of <0.05 will be considered statistically significant.

8. WORK PLAN: * **Months 1-2:** Software engineering of the anti-gravity simulation using Cursor/Antigravity; hardware integration.

- **Month 3:** Usability and safety testing on healthy volunteers; Institutional Ethics Committee clearance.
- **Months 4-6:** Patient recruitment, randomization, baseline assessments, and the 6-week clinical intervention phase.
- **Months 7-8:** Data analysis, tabulation of results, and drafting of the PG dissertation.

9. BUDGET: * AI Development Tools (Cursor/Antigravity Subscriptions): ~₹5,000

- Clinical Consumables (Disposable sEMG electrodes): ~₹15,000
- Printing, Documentation, and Contingency: ~₹5,000
- **Total Estimated Budget:** ₹25,000 (Self-funded).

10. ETHICAL ISSUES:

The study will adhere to ICMR ethical guidelines for biomedical research. IEC clearance will be obtained. Written informed consent will be acquired from all participants or legally authorized representatives. Conventional therapy will not be withheld from the control group. Participants in the experimental group will be monitored for VR-induced adverse effects.

11. BIOSAFETY ISSUES:

Standard clinical infection control protocols will be maintained. Single-use, disposable sEMG surface electrodes will be utilized for each participant. Hardware will be thoroughly sanitized with medical-grade wipes between patients.

12. UTILIZATION OF RESULTS:

This study will provide a pioneering proof-of-concept for the use of AI-assisted development tools by allied healthcare professionals to rapidly build custom rehabilitation software. Findings will advocate for the integration of highly customizable gamified biofeedback systems into neurogenic dysphagia protocols and will be published in peer-reviewed SLP journals.

13. REFERENCES:

1. Nordio S, Innocenti T, Agostini M, Meneghello F, Battel I. Biofeedback as an adjunctive treatment for post-stroke dysphagia: a pilot-randomized controlled trial. *Eur J Phys Rehabil Med*. 2022;58(2):226-233.

2. Liu Y, et al. Surface electromyographic biofeedback versus neuromuscular electrical stimulation for post-stroke dysphagia: a systematic review and network meta-analysis. *Front Neurol*. 2025.

3. Zhang L, et al. Effectiveness of Gamified Swallowing Exercises in Adults With Dysphagia: Systematic Review and Meta-Analysis of Randomized Controlled Trials. *JMIR Serious Games*. 2026;14:e82017.

4. Wu X, et al. The impact of VR technology based on swallowing auditory and visual stimulation on swallowing function and satisfaction in stroke patients with swallowing disorders. *Minerva Med*. 2024.

5. Crary MA, Mann GD, Groher ME. Initial psychometric assessment of a functional oral intake scale for dysphagia in stroke patients. *Arch Phys Med Rehabil*. 2005;86(8):1516-1520.

14. LIST OF ANNEXURES:

Annexure I: Participant Information Sheet (PIS)

Annexure II: Informed Consent Form (ICF)

Annexure III: The Functional Oral Intake Scale (FOIS)

Annexure IV: Swallowing Quality of Life Questionnaire (SWAL-QOL)

ANNEXURE I: PARTICIPANT INFORMATION SHEET (PIS)

Version # 1.0

Date: 29/04/2026

Short running title: AI-Assisted Custom VR Simulations for Neurogenic Dysphagia

Study Title: Effectiveness of AI-Developed (Cursor and Antigravity) Gamified Virtual Reality Simulations Using Surface Electromyography (sEMG) Biofeedback Compared to Conventional Therapy on Swallowing Function in Adults with Post-Stroke Neurogenic Dysphagia.

Investigators: Mr. Joju (Principal Investigator, 1st Year MSc SLP), [Your Name] (Guide), Dr. Anne Varghese (Co-Guide).

Department: Department of Speech-Language Pathology, Bangalore University.

Invitation to Participate You are invited to take part in a research study. Before you decide, please read this information carefully. This study is a research project and is not a part of your routine healthcare. You can take your time to decide and discuss this with your family.

1. What is the purpose of this study? The purpose of this study is to see if playing a computer game controlled by your swallowing muscles can help improve your swallowing faster than normal therapy. We have built a new digital game where your swallowing effort lifts a virtual object against gravity on a screen.

2. Why have I been chosen? You have been invited to participate because you have been diagnosed with swallowing difficulties (dysphagia) following a stroke. We are selecting a total of 30 participants for this study across Bangalore University clinics.

3. What will happen to me if I take part? If you agree to participate, the study will last for 6 weeks. You will be required to come for therapy 3 times a week, for 45 minutes each time. You will be placed into one of two groups by chance (like flipping a coin):

Group 1 (Normal Therapy): You will receive standard swallowing exercises taught by a therapist.

Group 2 (Game Therapy): You will do the same exercises, but we will place small, painless sticker sensors (sEMG) under your chin. These sensors read your muscle signals and send them to a computer. When you swallow hard, your muscle effort will control a game on the screen, like lifting an object against gravity. Before and after the 6 weeks, we

will ask you a few questions about your swallowing and rate your ability to eat and drink safely.

4. Do I have to take part? No, it is completely voluntary. If you refuse to participate, or decide to withdraw later, it will not affect your standard medical treatment, your care, or your legal rights in any way.

5. What are the possible risks and harms? There are no physical risks, pain, or medical side effects involved in this study. The sticker sensors placed under your chin are safe and painless. You might feel slightly tired from doing the swallowing exercises, just as you would in normal therapy.

6. What are the possible benefits? You may experience an improvement in your swallowing ability. The information we get from this study will help society and scientific advancement by showing if computer games can help stroke patients recover their swallowing faster.

7. Will I be compensated? No financial compensation will be provided for participating. However, the 6 weeks of swallowing therapy and the game-based treatment will be provided to you completely free of charge.

8. Will my information be kept confidential? Yes. Your privacy will be ensured during the study. All data will be kept strictly confidential, securely locked, and will only be accessible to the research team. Your name will be removed (anonymized) from any reports or publications that result from this study.

9. Who do I contact if I have questions?

If you have any doubts, you can contact:

- **Principal Investigator:** Mr. Joju, Phone: [Insert Number]
- **Ethics Committee Member-Secretary:** If you have concerns about your rights as a participant, you can contact the Bangalore University Ethics Committee at [Insert Phone/Email].

(Note: A translation of this document in Kannada will be provided to the participants).

ANNEXURE II: INFORMED CONSENT FORM (ICF)

Version # 1.0

Date: 29/04/2026

Short running title: AI-Assisted Custom VR Simulations for Neurogenic Dysphagia

Study title: Effectiveness of AI-Developed (Cursor and Antigravity) Gamified Virtual Reality Simulations Using Surface Electromyography (sEMG) Biofeedback Compared to Conventional Therapy on Swallowing Function in Adults with Post-Stroke Neurogenic Dysphagia

Protocol number: BU/SLP/2026/01

Names of all the research team members: Mr. Joju, Mr. Hemaraja Nayaka.S, Dr. Anne Varghese

Participant name: _____

Age: _____

Address: _____

Contact details:

Email: _____

Phone: _____

Name and address of the nominees and relation to participant:

_____

1. I have read (or have had read out) and understood the contents of the participant information sheet for the above mentioned study on (date)____ and I have been explained these details in my native tongue.
2. I have had ample opportunity and time to ask questions and clarify doubts from the research team whose contact details have been provided to me in the participant information sheet, in case of any further need.
3. I understand that my participation in the study is voluntary and that I am free to withdraw at any time, without giving any reason, without my medical care or legal rights as a patient, being affected. I hereby state that my decision to participate in this study is free from coercion or undue inducements.

4. I have been explained the purpose of the study and my responsibility in it. I have understood the possible risks and the benefits that might arise due to my enrollment.
5. I have been assured that my privacy will be respected and the data collected from me or my tissues will be kept confidentially secure.
6. I have also understood that the researchers might want to present the findings from the study or publish them in a scientific periodical or submit reports to the concerned authorities. I have been assured that in such situations my privacy and confidentiality will not be compromised.
7. I have also been informed that if my photographs are taken for the purpose of research, all efforts will be made to keep my identity confidential.
8. I understand that the sponsor/funding agency, others working on the sponsors' behalf, Ethics Committee, auditors/inspectors and representatives of the regulatory authorities will, at times, need to access my records collected for the purpose of this research and I hereby consent to the same. No one else shall be privy to my details without my explicit, prior permission.
9. I understand that as per the existing laws, the audio/audio-visual recording of my informed consent process will be done and I consent to the same.
10. I understand that my tissues and my data generated from this research study, will not be shared with any other researchers or be used in any other research study without my prior, explicit permission.
11. I have understood that my tissues and the data arising from this research will be securely stored for a period of 3 years and when disposed, will be done as per the biomedical waste disposal management policy of the institution.
12. I have been given to understand that none of the research team members have any conflict of interest arising out of this research study.
13. I have also understood that one copy of the informed consent document and one copy of the participant information sheet (in my native language) can be kept by me for future reference.

Participant's Signature / Name with Date:_____

Participant's thumb impression (in case illiterate): _____

PI's / trained research team member's Signature, Name with Date: _____

PI's contact details (Mobile/Email): _____

Independent witness signature, name with date: _____

ANNEXURE III: FUNCTIONAL ORAL INTAKE SCALE (FOIS)

Version # 1.0

Date: 29/04/2026

Short running title: AI-Assisted Custom VR Simulations for Neurogenic Dysphagia

Instructions: The FOIS is a 7-point clinician-rated scale that documents the functional oral intake of patients with dysphagia. Rate the patient's oral intake based on the following levels at Baseline and Post-Intervention.

- **Level 1:** Nothing by mouth (NPO).
- **Level 2:** Tube dependent with minimal attempts of food or liquid.
- **Level 3:** Tube dependent with consistent oral intake of food or liquid.
- **Level 4:** Total oral diet of a single consistency.
- **Level 5:** Total oral diet with multiple consistencies, but requiring special preparation or compensations.
- **Level 6:** Total oral diet with multiple consistencies without special preparation, but with specific food limitations.
- **Level 7:** Total oral diet with no restrictions.

ANNEXURE IV: SWALLOWING QUALITY OF LIFE QUESTIONNAIRE (SWAL-QOL)

Version # 1.0

Date: 29/04/2026

Short running title: AI-Assisted Custom VR Simulations for Neurogenic Dysphagia

Instructions for the Participant: Please read each statement carefully and circle the response that best describes your experience regarding your swallowing over the last month.

Note for Protocol: The full SWAL-QOL is a 44-item validated tool spanning 11 domains. Below is the structural representation of the questionnaire to be administered.

Scoring Likert Scale: 1 = Almost Always | 2 = Often | 3 = Sometimes | 4 = Hardly Ever | 5 = Never

Domains Assessed:

1. **Burden:** (e.g., "Dealing with my swallowing problem is very difficult.")
2. **Eating Duration:** (e.g., "It takes me longer to eat than other people.")
3. **Eating Desire:** (e.g., "I rarely feel hungry anymore.")
4. **Food Selection:** (e.g., "I have trouble figuring out what foods are safe for me to eat.")
5. **Communication:** (e.g., "People have trouble understanding me.")
6. **Fear:** (e.g., "I am afraid of choking when I eat.")
7. **Mental Health:** (e.g., "I feel depressed about my swallowing.")
8. **Social Functioning:** (e.g., "I do not go out to eat because of my swallowing problem.")
9. **Fatigue:** (e.g., "I feel weak.")
10. **Sleep:** (e.g., "I have trouble falling asleep.")
11. **Symptom Frequency:** (e.g., "Coughing", "Choking", "Food sticking in my throat" - Rated from 'Almost Always' to 'Never').